CALEB DOUGAL

Computer Science Student — Mathematics Minor

dougalcaleb.github.io | Provo, UT

Computer Science major (Mathematics minor) at Brigham Young University with hands-on experience across full-stack web development, cloud infrastructure, and independent systems engineering. Building toward graduate study and a career in quantum computing, with a parallel interest in backend and infrastructure engineering.

EDUCATION

Brigham Young University *Provo, UT*

B.S. Computer Science, Minor in Mathematics — Expected April 2028 | GPA: 3.8

Mountainland Technical College *Aug 2020 – May 2021*

Web Programming and Development — web technologies survey, software design principles

EXPERIENCE

Brigham Young University — Learning Suite Developer *Sept 2023 – Present*

- Develop full-stack features for BYU's Learning Suite platform using Vue, PHP, and MySQL, within a collaborative team of 5–10 developers
- Work across the stack from database schema to UI, following an Agile workflow with code review and shared ownership

Verisk — Software Contractor, Intern *Apr 2021 – Aug 2021*

- Built new features and resolved defects in a production Agile environment using JavaScript and Java
- Applied Git-based version control, unit testing, and relational database systems in a professional team setting

LEADERSHIP & ACTIVITIES

Quantum Information Association — Member *Present*

- Active member of BYU's quantum computing club; working with Qiskit outside coursework
- Helping organize the Intermountain Quantum Competition (Fall 2026), Organizing Committee

INDEPENDENT PROJECTS

Pylon Studios — Co-Founder & Lead Technical Architect *Ongoing*

- Building Pylon Fleet Manager, a cloud platform for competitive and hosted Terraria play, expanding an earlier solo project into a larger-scale, multi-tenant game hosting service
- Architecting a matchmaking and fleet-management backend on AWS GameLift (managed EC2 container fleets, Docker/ECR) supporting both short-lived challenge events and persistent, client-managed worlds, each with distinct AWS scaling paths

GitHub Wrapped — BYU HireReady Hackathon 2026

- Built a serverless pipeline (API Gateway, Python/TypeScript Lambdas, Aurora Serverless v2 via the RDS Data API, React/Vite on Amplify) that computes GitHub profile stats and uses AWS Bedrock (Claude) to generate a personalized slide deck for any handle

Parallel — CS260 Startup Project, BYU

- Built a full-stack social app surfacing real-world connections via photo-library location matching, on an AWS backend: Cognito, DynamoDB, S3, an EC2-hosted Node/Express API, and WebSockets for real-time nearby search

Additional: ongoing Unity game development (drag-and-drop systems, Addressables asset pipeline, pixel-perfect UI scaling) | Roundabout — open-source, documented vanilla JS carousel plugin

VOLUNTEER EXPERIENCE

Full-Time Missionary, The Church of Jesus Christ of Latter-day Saints — Tennessee Knoxville Mission (Aug 2021 – Aug 2023)

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C#, Java, PHP, Python

Infrastructure & Cloud: AWS (EC2, Global Accelerator, S3/CloudFront, GameLift), Linux/Bash, Nginx, Git, GitHub Actions (CI/CD)

Web & Data: Node.js/Express, Vue, HTML/CSS, MySQL and relational database design

Quantum: Qiskit, Amazon Braket

Other: Unity (game development), Agile/team development workflows